



MALAPPURAM SAHODAYA APTITUDE TEST 2024-25 DISTRICT LEVEL

Class: +2 PCM

Date: 04/01/2025

Marks: 100

Name of the Student: :.....

General Instructions:

1. Read the questions carefully.
2. Attempt all the questions.
3. No negative marking.
4. Answers to be marked in the OMR sheet. Darken the bubble completely. Partially darkened bubbles will not be considered.
5. Question number 1 to 100 carries 1 mark each.

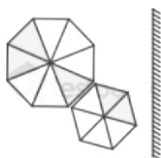
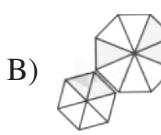
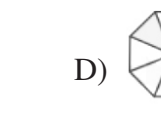
Read the passage given below and find the correct choices for questions 1-5 :

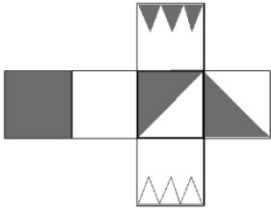
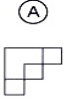
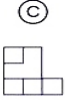
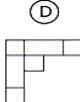
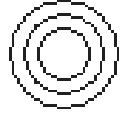
Climate change is one of the most pressing issues of our time. Rising temperatures, melting ice caps, and extreme weather events are all symptoms of a larger problem. The main culprit behind climate change is human activity, particularly the burning of fossil fuels such as coal, oil, and gas. The effects of climate change are far-reaching and devastating. Rising sea levels threaten coastal communities and cities, while droughts and heatwaves affect agriculture and food security. Climate change also has serious implications for human health, from heat stress and respiratory problems to the spread of disease. Despite the challenges, there are steps we can take to mitigate the effects of climate change. Transitioning to renewable energy sources, increasing energy efficiency, and protecting natural carbon sinks like forests are all critical. Governments, businesses, and individuals must work together to address this global problem.

On the basis of your understanding of the passage, answer the following questions given below by choosing the most appropriate option.

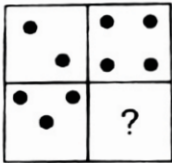
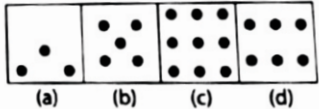
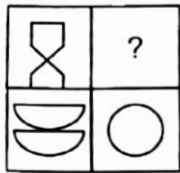
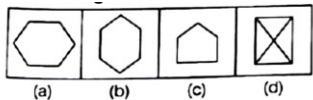
1. What is the author's tone in the passage?
A) Optimistic B) Pessimistic C) Neutral D) Sarcastic
2. According to the passage, what can be done to mitigate the effects of climate change?
A) Transitioning to renewable energy sources
B) Increasing energy efficiency
C) Protecting natural carbon sinks
D) All of the above
3. What is the author's purpose in writing the passage?
A) To persuade readers to take action on climate change
B) To inform readers about the effects of climate change
C) To entertain readers with a story about climate change
D) To analyze the economic impact of climate change
4. Which of the following sentences best summarizes the main idea of the passage?
A) Climate change is a natural phenomenon that cannot be stopped.
B) Human activity is the main cause of climate change.
C) The effects of climate change are limited to rising temperatures.
D) Climate change can be mitigated through individual actions.
5. What does the word "devastating" mean?
A) Causing great damage or destruction.
B) Causing minor damage or destruction.
C) Having no impact or effect.
D) Having a positive impact or effect.

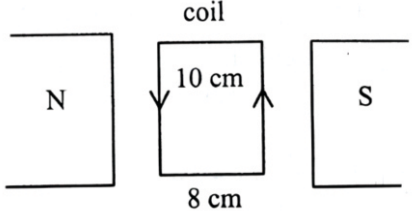
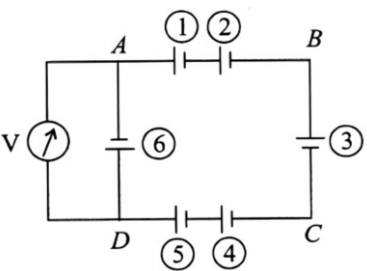
6.	Choose the correct meaning of the idiom from the given choices: To have an axe to grind A) A private end to serve B) To fail to arouse interest C) Personal reason or motivation for doing something. D) To work for both sides.
7.	Identify the literary device: Commencement of words with the same sound. A) Pun B) Alliteration C) Transferred epithet D) Oxymoron.
8.	Which term best describes a sad song. A) Ditty B) Knell C) Dirge D) Lay.
9.	Our task had been completed before sunset. Select the suitable active voice of the above statement. A) We completed our task before sunset. B) We have completed our task before sunset. C) We complete our task before sunset. D) We had completed our task before sunset.
10.	Has anybody answered your question? Select the suitable passive voice of the above statement. A) Your question has been answered? B) Anybody has answered your question? C) Has your question been answered? D) Have you answered your question?
11.	Choose the correct option from the given choices I said to him, " Why are you working so hard?" A) I asked him why he was working so hard. B) I asked him why was he working so hard. C) I asked him why had he been working so hard. D) I asked him why he had been working so hard.
12.	Fill in the blank with correct preposition. That child died _____ heavy fever. A) at B) for C) of D) with.
13.	Select the pair which has the same relationship AFTER: BEFORE A) First: Second B) Present: Past C) Contemporary: Historic D) Successor: Predecessor
14.	Choose the correct form of the verb from the given choices: If they _____ next to each other on the plane, they wouldn't have got married. A) hadn't sat B) had sat C) sat D) didn't sit.
15.	Choose the correct option from the bracket. The cake tasted _____ sweet that I couldn't finish it. A) so B) such C) very D) too
16.	Identify the literary device in this sentence. " My life is a dream " . A) Alliteration B) Simile C) Metaphor D) Personification.
17.	Which dramatic technique involves a character speaking their thoughts aloud directly to the audience, often revealing their inner turmoil.? A) Aside B) Soliloquy C) Monologue D) Dialogue.
18.	A Haiku is a _____ poetic form. A) two- line B) three - line C) four - line D) six - line.

19.	A poem that reflects upon death or loss is _____. A) Epic B) Pastoral Poetry C) Elegy D) Ode.
20.	Which of the following sentences is grammatically correct? A) Me and my friend went to the park. B) My friend and I went to the park. C) I and my friend went to the park. D) My friend and me went to the park.
21.	Choose the correct option from the given choices. Although he was _____, he remained optimistic about the future. A) Wealthy B) Destitute C) Jubilant D) Fortunate
22.	Read the following statements and fill in the gaps using correct question tags. We often go out for dinner on weekends, _____ ? A) Don't we? B) Do we? C) Does we? D) Doesn't we?
23.	Choose the correct meaning of idiom "To face the music " means _____ A) to be greeted rudely. B) to be offered warm hospitality. C) to be on the front seat in a music concert. D) to bear the consequences.
24.	Choose the correct option from the given choices: We _____ off old clothes and bought new ones for Diwali. A) cost B) coast C) caste D) cast
25.	Fill in the blank using the correct form of the verb given in the bracket. By this time next year, I _____ (take) all my exams. A) will take B) will have taken C) have taken D) take.
26.	If the symbols followed by consonants interchange their positions within the group, then which element is third from the right end? B @ C 7 N R % 5 \$ G 6 K M & 4 S # P U 5 A) U B) # C) P D) 5
27.	Gaurav walks 20 metres towards North. He then turns left and walks 40 metres. He again turns left and walks 20 metres. Further, he moves 20 metres after turning to the right. How far is he from his original position? A) 20 metres B) 30 metres C) 50 metres D) 60 metres
28.	 For the picture shown above which one of the following is the correct picture representing reflection with respect to the mirror shown as the dotted line A)  B)  C)  D) 

29.	Which of the four possible options represents the cube in its folded form?  A  B  C  D  E 
30.	What would the 3D shape look like from above?  A) A C) C  B) B D) D  C) C  D) D 
31.	Errata: Books : : Flaws : ? A) Manuscripts B) Metals C) Speech D) Charter
32.	Earthquake is to Tremor as Thunder is to _____. A) Lightning B) Storm C) Roar D) Strike
33.	Paw : Cat :: Hoof : ? A) Lamb B) Elephant C) Lion D) Horse
34.	Look at this series: 14, 28, 20, 40, 32, 64, ... What number should come next? A) 52 B) 56 C) 96 D) 128
35.	Look at this series: 8, 6, 9, 23, 87, ... What number should come next? A) 128 B) 226 C) 324 D) 429
36.	Which of the following diagrams indicates the best relation between Women, Mothers and Engineers ? A)  B)  C)  D) 
37.	Complete the pattern: FAG, GAF, HAI, IAH, _____ A) JAK B) HAL C) HAK D) JAI
38.	DEF, DEF2, DE2F2, _____, D2E2F3 A) DEF3 B) D3EF3 C) D2E3F D) D2E2F2
39.	Play is to actor as concert is to A) Symphony B) Musician C) Piano D) Percussion

40.	Here are some words translated from an artificial language. moolokarn means blue sky wilkospadi means bicycle race moolowilko means blue bicycle Which word could mean "racecar"? A) Wilkozwet B) Spadiwilko C) Moolobreil D) Spadivolo									
41.	Statements: In a one-day cricket match, the total runs made by a team were 200. Out of these 160 runs were made by spinners. Conclusions: 1. 80% of the team consists of spinners. 2. The opening batsmen were spinners. A) Only conclusion I follows B) Only conclusion II follows C) Either I or II follows D) Neither I nor II follows									
42.	A, B, C, D and E are sitting on a bench. A is sitting next to B, C is sitting next to D, D is not sitting with E who is on the left end of the bench. C is on the second position from the right. A is to the right of B and E. A and C are sitting together. In which position is A sitting? A) Between B and D B) Between B and C C) Between E and D D) Between C and E									
43.	If A + B means A is the mother of B; A - B means A is the brother B; A % B means A is the father of B and A x B means A is the sister of B, which of the following shows that P is the maternal uncle of Q? A) Q - N + M x P B) P + S x N - Q C) P - M + N x Q D) Q - S % P									
44.	If P \$ Q means P is the brother of Q; P # Q means P is the mother of Q; P * Q means P is the daughter of Q in A # B \$ C * D, who is the father? A) D B) B C) C D) Data is inadequate									
45.	In a certain code 'MISSIONS' is written as 'MSIISNOS'. How is 'ONLINE' written in that code? A) OLNIE B) ONILEN C) NOILEN D) LNOENI									
46.	In certain code 'FROZEN' is written as 'OFAPSG'. Then how would 'MOLTEN' be written in that code? A) OFPOMN B) OFSMPN C) OFUMPN D) OFUMON									
47.	Select a suitable figure from the four alternatives that would complete the figure matrix. <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td>?</td></tr></table> (1) (2) (3) (4) A) 1 B) 2 C) 3 D) 4									?
		?								
48.	Select a suitable figure from the four alternatives that would complete the figure matrix. <table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td>?</td></tr></table> (1) (2) (3) (4) A) 1 B) 2 C) 3 D) 4									?
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49.	In the following question, find out which of the answer figures (a), (b), (c) and (d) completes the figure matrix? Problem Figure:  Answer Figures:  A) (a) B) (b) C) (c) D) (d)
50.	In the following question, find out which of the answer figures (a), (b), (c) and (d) completes the figure matrix? Problem Figure:  Answer figures:  A) (a) B) (b) C) (c) D) (d)
51.	Three charges $+Q, q, +Q$ are placed respectively, at distance $0, d/2$ and d from the origin, on the x-axis. If the net force experienced by $+Q$ placed at $x=0$ is zero, then value of q is A) $+Q/4$ B) $-Q/2$ C) $+Q/2$ D) $-Q/4$
52.	In an interference experiment the ratio of amplitudes of coherent waves is $(a_1/a_2)=(1/3)$. The ratio of maximum and minimum intensities of fringes will be A) 4 : 1 B) 9 : 1 C) 2 : 1 D) 18 : 1
53.	A magnetic needle of magnetic moment 6.7×10^{-2} Am² and moment of inertia 7.5×10^{-6} kgm² is performing simple harmonic oscillations in a magnetic field of 0.01 T . Time taken for 10 complete oscillations is A) 6.65 s B) 8.89 s C) 6.98 s D) 8.76 s
54.	A parallel plate capacitor has $1\mu\text{F}$ capacitance. One of its two plates is given $+2\mu\text{C}$ charge and the other plate, $+4\mu\text{C}$ charge. The potential difference developed across the capacitor is A) 3 V B) 2 V C) 5 V D) 1 V
55.	Photon of frequency f has a momentum associated with it. If c is the velocity of light, the momentum is A) f/c B) hf/c C) hf/c^2 D) hfc
56.	A convergent doublet of separated lenses, corrected for spherical aberration, has a resultant focal length of 10 cm . The separation between the two lenses is 2 cm . The focal lengths of the component lenses are A) 10cm, 12cm B) 16cm, 18cm C) 12cm, 14cm D) 18cm, 20cm
57.	A nucleus disintegrates into two nuclear parts which have their velocities in the ratio 2:1. The ratio of their nuclear sizes will be A) $2^{1/3}:1$ B) $1:2^{1/3}$ C) $3^{1/2}:1$ D) $1:3^{1/2}$

58.	A coil of 50 turns carrying a current of 2 A in a magnetic field of 0.5 T. The torque acting on the coil is  coil 10 cm 8 cm A) 0.4 Nm clockwise B) 0.8 Nm anticlockwise C) 0.4 Nm anticlockwise D) 0.8 Nm anticlockwise
59.	The shortest wavelength of Paschen series in hydrogen spectrum is 8182. The first member of the Paschen series is nearly A) 13400 B) 15400 C) 16700 D) 18700
60.	Six cells, each of emf 5 V and internal resistance 0.1Ω are connected as shown in Figure. The reading of the ideal voltmeter V is  A) 0 V B) 15 V C) 5 V D) 30 V
61.	Two equal resistances when connected in series to a battery consume electric power of 60 W . If these resistances are now connected in parallel combination to the same battery, the electric power consumed will be A) 30 W B) 60 W C) 120 W D) 240 W
62.	At absolute zero, silicon (si) acts as A) non-metal B) metal C) insulator D) none of these
63.	Two coaxial solenoids are made by winding thin insulated wire over a pipe of cross-sectional area $A=10\text{cm}^2$ and length $=20\text{cm}$. If one of the solenoids has 300 turns and the other 400 turns, their mutual inductance is $\mu_0=4\pi\times 10^{-7}\text{TmA}^{-1}$ A) $4.8\pi\times 10^{-4}\text{H}$ B) $4.8\pi\times 10^{-5}\text{H}$ C) $2.4\pi\times 10^{-4}\text{H}$ D) $2.4\pi\times 10^{-5}\text{H}$
64.	Assume that a radio station is about 200 km away from your location and the station operates at 972 kHz . How long does it take for an electromagnetic signal to travel from the station to you and how many wave crests does it send out per second A) $555\mu\text{s}$ and 0.972×10^5 crests per second B) $555\mu\text{s}$ and 97.2×10^7 crests per second C) $666\mu\text{s}$ and 9.72×10^5 crests per second D) $666\mu\text{s}$ and 972×10^5 crests per second
65.	The relation between the charge flow ΔQ through the circuit of resistance r and the change in the magnetic flux $\Delta\phi_B$ is A) $\Delta Q = \frac{\Delta\phi_B}{r}$ B) $\Delta Q = \frac{r}{\phi_B}$ C) $\Delta\phi_B = \frac{\Delta Q}{r}$ D) $\Delta\phi_B = \frac{\Delta Q}{r^2}$
66.	In the depression of freezing point experiment A) vapour pressure of the solution is less than that of pure solvent B) vapour pressure of the solution is more than that of pure solvent C) only solute molecules solidify at the freezing point D) only solvent molecules solidify at the freezing point.

	Choose the most appropriate answer from the options given below. A) A only B) B and C only C) A and C only D) A and D only
67.	The equation that is incorrect is A) $(\Delta_m^\circ)_{\text{H}_2\text{O}} = (\Delta_m^\circ)_{\text{HCl}} + (\Delta_m^\circ)_{\text{NaOH}} - (\Delta_m^\circ)_{\text{NaCl}}$ B) $(\Delta_m^\circ)_{\text{NaBr}} - (\Delta_m^\circ)_{\text{NaI}} = (\Delta_m^\circ)_{\text{KBr}} - (\Delta_m^\circ)_{\text{NaBr}}$ C) $(\Delta_m^\circ)_{\text{NaBr}} - (\Delta_m^\circ)_{\text{NaCl}} = (\Delta_m^\circ)_{\text{KBr}} - (\Delta_m^\circ)_{\text{KCl}}$ D) $(\Delta_m^\circ)_{\text{KCl}} - (\Delta_m^\circ)_{\text{NaCl}} = (\Delta_m^\circ)_{\text{KBr}} - (\Delta_m^\circ)_{\text{NaBr}}$
68.	If 50 % of a reaction occurs in 100 second and 75% of the reaction occurs in 200 second, the order of this reaction is A) 1 B) 2 C) zero D) 3
69.	For a reaction of order n, the unit of the rate constant is A) $\text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$ B) $\text{mol}^{1-n} \text{L}^{1-n} \text{s}^{-1}$ C) $\text{mol}^{1-n} \text{L}^{1-n} \text{s}$ D) $\text{mol}^{1-n} \text{L}^{2n} \text{s}^{-1}$
70.	What is the magnetic moment for Mn 2+ ion? A) 5.92 B.M. B) 2.80 B.M. C) 8.95 B.M. D) 9.55 B.M.
71.	Indicate the complex / complex ion which did not show any geometrical isomerism. A) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ B) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$ C) $[\text{Co}(\text{CN})_5(\text{NC})]^{3-}$ D) $[\text{CoCl}_2(\text{en})_2]$
72.	The compound C_7H_8 undergoes the following reactions: $\text{C}_7\text{H}_8 \xrightarrow{3\text{Cl}_2/\Delta} \text{A} \xrightarrow{\text{Br}_2/\text{Fe}} \text{B} \xrightarrow{\text{Zn}/\text{HCl}} \text{C}$ The product C is A) <i>m</i>-bromotoluene B) <i>o</i>-bromotoluene C) 3-bromo-2,4,6-trichlorotoluene D) <i>p</i>-bromotoluene.
73.	Incorrect method of preparation for alcohols from the following is A) reaction of ketone with RMgBr followed by hydrolysis B) hydroboration-oxidation of alkene C) ozonolysis of alkene D) reaction of alkyl halide with aqueous NaOH.
74.	In Reimer-Tiemann reaction, the reagent used is A) $\text{CHCl}_3, \text{aq. NaOH}$ B) $[\text{O}], \text{I}_2/\text{NaOH}$ C) $\text{NaOH}, \text{CO}_2, \text{H}^+$ D) $\text{NaHCO}_3, \text{Cu}, \Delta$
75.	Reaction between benzaldehyde and acetophenone in presence of dilute NaOH is known as A) Aldol condensation B) Cannizzaro's reaction C) Cross Cannizzaro's reaction D) Cross Aldol condensation.
76.	The major product of the following reaction is $\text{R}-\text{C}=\text{N}$ $\xrightarrow[\text{(2)H}_2\text{O}]{\text{(1)AlH(i-Bu)}_2}$ A) RCOOH C) RCONH_2 B) RCHO D) RCH_2NH_2

77.	Carboxylic acids have higher boiling points than aldehydes, ketones and even alcohols of comparable molecular mass. It is due to their A) formation of intramolecular H-bonding B) formation of carboxylate ion C) more extensive association of carboxylic acid via van der Waals' forces of attraction D) formation of intermolecular H-bonding
78.	The reaction, $\frac{1}{2} \text{H}_{2(g)} + \text{AgCl}_{(s)} \rightleftharpoons \text{H}^+_{(aq)} + \text{Cl}^-_{(aq)} + \text{Ag}_{(s)}$ occurs in which of the given galvanic cell? A) $\text{Ag} \text{AgCl}_{(s)} \text{KCl}_{(\text{sol}^n)} \text{AgNO}_3 \text{Ag}$ B) $\text{Pt} \text{H}_{2(g)} \text{HCl}_{(\text{sol}^n)} \text{AgNO}_3(\text{sol}^m) \text{Ag}$ C) $\text{Pt} \text{H}_{2(g)} \text{KCl}_{(\text{sol}^n)} \text{AgCl}_{(s)} \text{Ag}$ D) $\text{Pt} \text{H}_{2(g)} \text{HCl}_{(\text{sol}^n)} \text{AgCl}_{(s)} \text{Ag}$
79.	Which of the following amines can be prepared by Gabriel phthalimide reaction? A) Triethylamine B) n-Butylamine C) t-Butylamine D) neo-Pentylamine
80.	Among the following compounds most basic amino acid is A) histidine B) serine C) asparagine D) lysine.
81.	If f is a real valued function such that $f(x+y)=f(x)+f(y)$ and $f(1)=5$, then the value of $f(100)$ is A) 200 B) 300 C) 350 D) 500
82.	If $\tan^{-1}(x+2) + \tan^{-1}(x-2) - \tan^{-1}\left(\frac{1}{2}\right) = 0$, then one of the values of x is A) -1 B) 5 C) $\left(\frac{1}{2}\right)$ D) 1
83.	83. If $\sin^{-1}\left(\frac{x}{5}\right) + \operatorname{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$, then the value of x is A) 1 B) 3 C) 4 D) 5
84.	If $A = \begin{bmatrix} 2-k & 2 \\ 1 & 3-k \end{bmatrix}$ is a singular matrix, then the value of $5k-k^2$ is A) 6 B) -6 C) 4 D) -4
85.	If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $ A =4$, then α is equal to A) 4 B) 11 C) 5 D) 0
86.	If the derivative of the function f(x) is everywhere continuous and is given by $f(x) = \begin{cases} bx^2 + ax + 4, & x \geq -1 \\ ax^2 + b, & x < -1 \end{cases}$, then A) $a=2, b=-3$ B) $a=3, b=2$ C) $a=-2, b=-3$ D) $a=-3, b=-2$
87.	If $y = \tan^{-1} \sqrt{\frac{1-\sin x}{1+\sin x}}$, then the value of $\frac{dy}{dx}$ at $x = \frac{\pi}{6}$ is A) $-\frac{1}{2}$ B) $\frac{1}{2}$ C) 1 D) -1
88.	A ladder 10 m long rests against a vertical wall with the lower end on the horizontal ground. The lower end of the ladder is pulled along the ground away from the wall at the rate of 3 cm/s. The height of the upper end while it is descending at the rate of 4 cm/s, is A) $4\sqrt{3}$ m B) $5\sqrt{3}$ m C) 8 m D) 6 m

89.	The slope of the normal to the curve $y = x^2 - \frac{1}{x^2}$ at $(-1, 0)$ is A) $\frac{1}{4}$ B) $-\frac{1}{4}$ C) 4 D) -4
90.	$\int \left(1 + x - \frac{1}{x}\right) e^{x+\frac{1}{x}} dx$ is equal to A) $(x-1)e^{x+\frac{1}{x}} + C$ B) $xe^{x+\frac{1}{x}} + C$ C) $(x+1)e^{x+\frac{1}{x}} + C$ D) $-xe^{x+\frac{1}{x}} + C$
91.	The value of $\int_1^e \frac{1+\log x}{3x} dx$ is A) $\frac{1}{4}$ B) $\frac{1}{2}$ C) $\frac{3}{4}$ D) e
92.	Area of the region bounded by two parabolas $y=x^2$ and $x=y^2$ is A) $\frac{1}{4}$ B) 4 C) $\frac{1}{3}$ D) 3
93.	The area bounded by $y = \sin^{-1} x$, $x = \frac{1}{\sqrt{2}}$ and X-axis is A) $\left(\frac{1}{\sqrt{2}} + 1\right) sq\ units$ B) $\left(1 - \frac{1}{\sqrt{2}}\right) sq\ units$ C) $\frac{\pi}{4\sqrt{2}} sq\ units$ D) $\left(\frac{\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}} - 1\right) sq\ units$
94.	The order and degree of the differential equation $\sqrt{\sin x} (dx + dy) = \sqrt{\cos x} (dx - dy)$ are A) (1,2) B) (2,2) C) (1,1) D) (2,1)
95.	The solution of the differential equation $\frac{dy}{dx} = e^{y+x} + e^{y-x}$ is A) $e^{-y} = e^x - e^{-x} + C$ B) $e^{-y} = e^{-x} - e^x + C$ C) $e^{-y} = e^x + e^{-x} + C$ D) $e^{-y} + e^x + e^{-x} = C$
96.	If the scalar product of the vector $\hat{i} + \hat{j} + 2\hat{k}$ with the unit vector along $m\hat{i} + 2\hat{j} + 3\hat{k}$ is equal to 2, then one of the values of m is A) 3 B) 4 C) 5 D) 6
97.	If $a = \hat{i} + 2\hat{j} + 2\hat{k}$, $ b = 5$ and the angle between a and b is $\pi/6$, then the area of the triangle formed by these two vectors as two sides is A) $\frac{15}{4}$ B) $\frac{15}{2}$ C) 15 D) $\frac{15\sqrt{3}}{2}$
98.	If the lines $\frac{1-x}{3} = \frac{y-2}{2\alpha} = \frac{z-3}{2}$ and $\frac{x-1}{3\alpha} = y-1 = \frac{6-z}{5}$ are perpendicular, then the value of α is A) $\frac{10}{7}$ B) $-\frac{10}{7}$ C) $-\frac{10}{11}$ D) $\frac{10}{11}$
99.	If given constraints are $5x+4y \geq 2$, $x \leq 6$ and $y \leq 7$ then the maximum value of the function $z=x+2y$ is A) 13 B) 14 C) 15 D) 20
100.	A bag contains 3 black, 3 white and 2 red balls. One by one, three balls are drawn without replacement. The probability that the third ball is red, is A) $\frac{2}{56}$ B) $\frac{14}{56}$ C) $\frac{12}{56}$ D) $\frac{3}{56}$